

✓ HW1: The Normal Distribution

(a) Key characteristics of the Normal distribution

The Normal (Gaussian) distribution is a continuous probability distribution whose graph is a symmetric, bell-shaped curve. Its key characteristics are:

- **Symmetry:** The curve is perfectly symmetric about its center, so the mean, median, and mode are all equal.
- **Bell shape:** Probability density is highest at the center and decreases smoothly toward both tails, which extend infinitely in both directions without ever touching zero (asymptotic tails).
- **Defined by two parameters:** A Normal distribution is fully described by just its mean (μ) and standard deviation (σ), written $X \sim N(\mu, \sigma^2)$.
- **Total area equals 1:** Like all probability distributions, the total area under the curve equals 1 (100% of outcomes).

Role of μ (mean): The mean determines the **location** (center) of the distribution on the number line. Changing μ shifts the entire bell curve left or right without altering its shape.

Role of σ (standard deviation): The standard deviation determines the **spread/width** (shape) of the distribution. A small σ produces a tall, narrow curve (values tightly clustered around the mean), while a large σ produces a short, wide curve (values more spread out). σ does not affect the center of the distribution.

(b) The Empirical Rule (68-95-99.7 rule)

For any Normal distribution, the Empirical Rule states that data falls within a predictable range of standard deviations from the mean:

- Approximately **68%** of observations lie within **1 standard deviation** of the mean: $[\mu - \sigma, \mu + \sigma]$
- Approximately **95%** of observations lie within **2 standard deviations** of the mean: $[\mu - 2\sigma, \mu + 2\sigma]$
- Approximately **99.7%** of observations lie within **3 standard deviations** of the mean: $[\mu - 3\sigma, \mu + 3\sigma]$

This rule gives a quick, practical way to gauge how likely or unusual a given observation is, and it underlies why observations beyond 3 standard deviations are considered rare outliers.

(c) Probability that inflation is between 1.5 and 6

Given $X \sim N(\mu = 3, \sigma = 1.5)$, we want $P(1.5 \leq X \leq 6)$.

Convert each boundary to a Z-score using $Z = \frac{X - \mu}{\sigma}$:

$$Z_1 = \frac{1.5 - 3}{1.5} = \frac{-1.5}{1.5} = -1.0$$
$$Z_2 = \frac{6 - 3}{1.5} = \frac{3}{1.5} = 2.0$$

So we need $P(-1.0 \leq Z \leq 2.0) = \Phi(2.0) - \Phi(-1.0)$, where Φ is the standard Normal CDF.

Using standard Normal tables: $\Phi(2.0) \approx 0.9772$ and $\Phi(-1.0) \approx 0.1587$.

$$P(-1.0 \leq Z \leq 2.0) \approx 0.9772 - 0.1587 = 0.8186$$

So there is approximately an 81.9% probability that the inflation rate in a randomly selected year falls between 1.5% and 6%.

```
1 from scipy.stats import norm
2
3 mu = 3
4 sigma = 1.5
5
6 x1, x2 = 1.5, 6
7
8 z1 = (x1 - mu) / sigma
9 z2 = (x2 - mu) / sigma
10
11 prob = norm.cdf(z2) - norm.cdf(z1)
12
13 print(f"Z1 = {z1:.4f}")
14 print(f"Z2 = {z2:.4f}")
15 print(f"P({x1} <= X <= {x2}) = P({z1:.2f} <= Z <= {z2:.2f}) = {prob:.4f} ({prob*100:.2f}%)")
16
```



```
Z1 = -1.0000
Z2 = 2.0000
```

Question 2: Exploring Income Distribution

Household income $X \sim N(\mu = 70000, \sigma = 18000)$.

(a) Percentage of households with income between 52,000 and 106,000

Convert boundaries to Z-scores using $Z = \frac{X - \mu}{\sigma}$:

$$Z_1 = \frac{52000 - 70000}{18000} = \frac{-18000}{18000} = -1.0$$

$$Z_2 = \frac{106000 - 70000}{18000} = \frac{36000}{18000} = 2.0$$

So we need $P(-1.0 \leq Z \leq 2.0) = \Phi(2.0) - \Phi(-1.0) \approx 0.9772 - 0.1587 = 0.8186$.

So approximately 81.9% of households have an income between 52,000 and 106,000.

(b) Minimum income to be in the top 5% of earners

The top 5% corresponds to the 95th percentile, whose Z-score is $Z_{0.95} \approx 1.645$.

$$X = \mu + Z \cdot \sigma = 70000 + 1.645 \times 18000 \approx 70000 + 29610 = 99610$$

A household needs to earn at least approximately \$99,610 to be in the top 5% of earners.

(c) Maximum income to qualify for the bottom-20% assistance program

The bottom 20% corresponds to the 20th percentile, whose Z-score is $Z_{0.20} \approx -0.8416$.

$$X = \mu + Z \cdot \sigma = 70000 + (-0.8416) \times 18000 \approx 70000 - 15149 = 54851$$

A household can earn at most approximately \$54,851 to qualify for the program.

```
1 import numpy as np
2 from scipy.stats import norm
3 mu_income = 70000
4 sigma_income = 18000
5
6 # (a) P(52000 <= X <= 106000)
7 x1_income, x2_income = 52000, 106000
8 z1_income = (x1_income - mu_income) / sigma_income
9 z2_income = (x2_income - mu_income) / sigma_income
10 prob_income = norm.cdf(z2_income) - norm.cdf(z1_income)
11
12 print(f"(a) Z1 = {z1_income:.4f}, Z2 = {z2_income:.4f}")
13 print(f"(a) P({x1_income} <= X <= {x2_income}) = {prob_income:.4f} ({prob_income*100:.2f}%)")
14
15 # (b) Top 5% => 95th percentile
16 z_top5 = norm.ppf(0.95)
17 x_top5 = mu_income + z_top5 * sigma_income
18 print(f"(b) Z(0.95) = {z_top5:.4f}, minimum income for top 5% = ${x_top5:,.2f}")
19
20 # (c) Bottom 20% => 20th percentile
21 z_bottom20 = norm.ppf(0.20)
22 x_bottom20 = mu_income + z_bottom20 * sigma_income
23 print(f"(c) Z(0.20) = {z_bottom20:.4f}, maximum income for bottom 20% = ${x_bottom20:,.2f}")
24
```

```
(a) Z1 = -1.0000, Z2 = 2.0000
(a) P(52000 <= X <= 106000) = 0.8186 (81.86%)
(b) Z(0.95) = 1.6449, minimum income for top 5% = $99,607.37
(c) Z(0.20) = -0.8416, maximum income for bottom 20% = $54,850.82
```

Question 3: Exploring Cars Dataset

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4 import numpy as np
5 import scipy.stats as stats
```

```

6 import warnings
7 warnings.filterwarnings("ignore")
8
9
10 # import cars dataset
11 import statsmodels.api as sm
12
13 cars = sm.datasets.get_rdataset("cars", "datasets").data
14
15 cars_df = pd.DataFrame(cars)
16 print(cars_df.head())

```

```

   speed  dist
0      4     2
1      4    10
2      7     4
3      7    22
4      8    16

```

```
1 print(cars_df.describe())
```

```

count    speed      dist
mean    15.400000  42.980000
std      5.287644  25.769377
min      4.000000   2.000000
25%     12.000000  26.000000
50%     15.000000  36.000000
75%     19.000000  56.000000
max     25.000000 120.000000

```

```

1 plt.scatter(cars_df["speed"], cars_df["dist"])
2 plt.title("Scatter plot of Speed vs Stopping Distance")
3 plt.xlabel("Speed (mph)")
4 plt.ylabel("Stopping Distance (ft)")
5 plt.tight_layout()
6 plt.title("Scatter plot of Speed vs Stopping Distance")
7 plt.show()

```



```
1 print(cars_df[["speed", "dist"]].corr())
```

```

      speed      dist
speed  1.000000  0.806895
dist   0.806895  1.000000

```

Positive relationship between speed and stopping distance is expected because as a car's speed increases the distance required to stop also increases due to greater momentum and kinetic energy

```

1
2 # (a) Summary statistics for dist

```

```

3 dist_mean = cars_df["dist"].mean()
4 dist_median = cars_df["dist"].median()
5 dist_std = cars_df["dist"].std()
6 dist_min = cars_df["dist"].min()
7 dist_max = cars_df["dist"].max()
8
9 print("(a) Summary statistics for dist:")
10 print(f"    mean  = {dist_mean:.2f}")
11 print(f"    median = {dist_median:.2f}")
12 print(f"    std   = {dist_std:.2f}")
13 print(f"    min   = {dist_min}")
14 print(f"    max   = {dist_max}")
15
16 # (b) Histogram of dist
17 plt.figure(figsize=(7, 5))
18 plt.hist(cars_df["dist"], bins=10, color="steelblue", edgecolor="black")
19 plt.title("Histogram of Car Stopping Distances")
20 plt.xlabel("Stopping Distance (ft)")
21 plt.ylabel("Frequency")
22 plt.tight_layout()
23 plt.show()

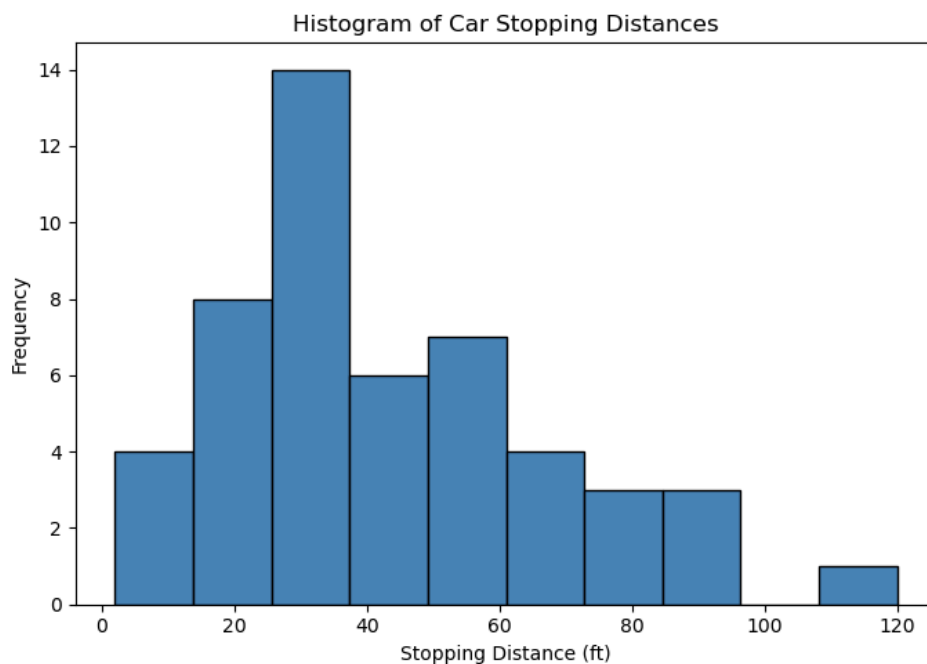
```

(a) Summary statistics for dist:

```

mean  = 42.98
median = 36.00
std    = 25.77
min    = 2
max    = 120

```



```

1 import statsmodels.api as sm
2
3 X = sm.add_constant(cars_df["speed"])
4 y = cars_df["dist"]
5 model = sm.OLS(y, X).fit()
6 print(model.summary())

```

OLS Regression Results

```

=====
Dep. Variable:          dist    R-squared:                0.651
Model:                  OLS    Adj. R-squared:           0.644
Method:                 Least Squares    F-statistic:         89.57
Date:                   Mon, 31 Aug 2026    Prob (F-statistic):   1.49e-12
Time:                   20:22:53    Log-Likelihood:       -206.58
No. Observations:       50    AIC:                  417.2
Df Residuals:           48    BIC:                  421.0
Df Model:               1
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	-17.5791	6.758	-2.601	0.012	-31.168	-3.990
speed	3.9324	0.416	9.464	0.000	3.097	4.768

Omnibus:	8.975	Durbin-Watson:	1.676
Prob(Omnibus):	0.011	Jarque-Bera (JB):	8.189
Skew:	0.885	Prob(JB):	0.0167
Kurtosis:	3.893	Cond. No.	50.7

=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

(c) Interpreting the regression model

The fitted model is $\text{dist} \approx -17.58 + 3.93 \times \text{speed}$. The slope (≈ 3.93) is statistically significant ($p < 0.001$), meaning each additional mph of speed adds about 3.93 ft to stopping distance, on average. The model explains about 65% of the variation in stopping distance ($R^2 \approx 0.65$), confirming the strong positive relationship seen in the scatter plot and correlation.

Question 4: The Central Limit Theorem (CLT)

(a) What does the CLT say?

The Central Limit Theorem says that if you repeatedly take random samples of a sufficiently large size (n) from *any* population — regardless of the shape of that population's distribution — the distribution of the **sample means** will be approximately **Normally distributed**, centered at the population mean μ , with the spread of sample means shrinking as n grows. In short: individual data can look messy or skewed, but averages of that data behave like a bell curve.

(b) Conditions under which the CLT generally holds

1. **Random sampling:** Samples must be drawn randomly (and independently) from the population.
2. **Sufficiently large sample size:** n should be large enough (commonly $n \geq 30$ is used as a rule of thumb) — the more skewed the population, the larger n needs to be.
3. **Finite variance:** The population must have a finite mean μ and finite standard deviation σ (no infinite/undefined variance).

(c) Sampling distribution of mean productivity ($n = 100$) from a right-skewed population

Even though the population of individual worker productivity is highly right-skewed, taking many large samples of $n = 100$ means the **sampling distribution of the sample mean will be approximately Normal (bell-shaped)** — the CLT "averages out" the skew from the superstar workers because extreme individual values get diluted when averaged over 100 workers.

Its **mean** will equal the population mean: $\mu_{\bar{x}} = \mu$.

Its **standard deviation** (the standard error) will be much smaller than the population's, shrinking with sample size:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{\sqrt{100}} = \frac{\sigma}{10}$$

So the sampling distribution is centered at the same value as the population but is far more tightly clustered and symmetric — a bell curve, not a skewed shape.

Question 5: Simulating the CLT with the `iris` Sepal.Length variable

```
1 # import iris dataset
2
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5
6 iris = sns.load_dataset("iris")
```

```
1 print(iris.head())
2 print(iris.describe())
3 print(iris.info())
4 print(iris.shape)
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
	sepal_length	sepal_width	petal_length	petal_width	
count	150.000000	150.000000	150.000000	150.000000	

```

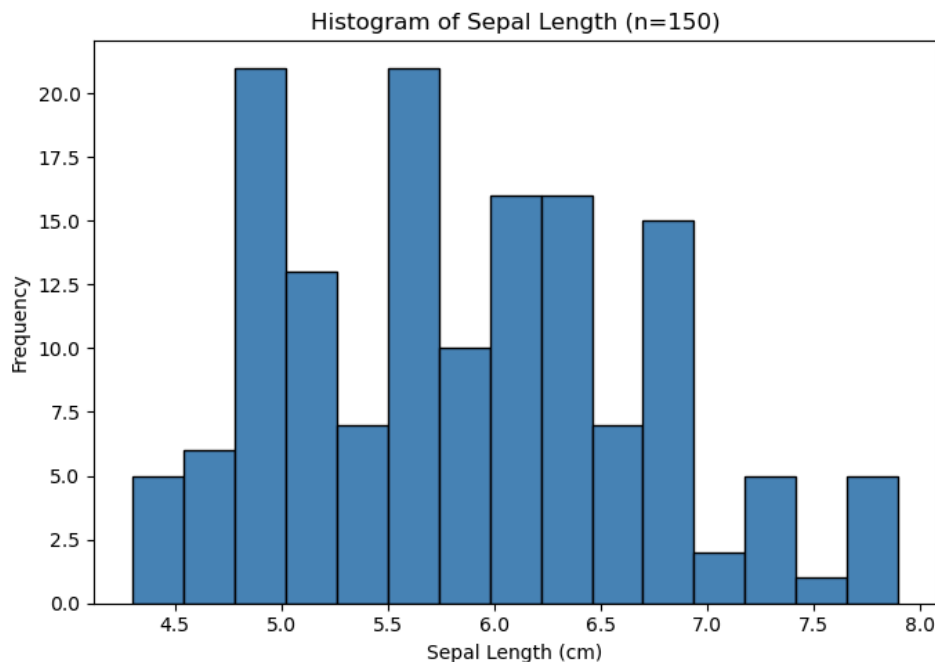
mean      5.843333    3.057333    3.758000    1.199333
std       0.828066    0.435866    1.765298    0.762238
min       4.300000    2.000000    1.000000    0.100000
25%      5.100000    2.800000    1.600000    0.300000
50%      5.800000    3.000000    4.350000    1.300000
75%      6.400000    3.300000    5.100000    1.800000
max       7.900000    4.400000    6.900000    2.500000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
(150, 5)

```

```

1 # (a) Histogram of Sepal.Length for all 150 flowers, population mean and std
2 plt.figure(figsize=(7, 5))
3 plt.hist(iris["sepal_length"], bins=15, color="steelblue", edgecolor="black")
4 plt.title("Histogram of Sepal Length (n=150)")
5 plt.xlabel("Sepal Length (cm)")
6 plt.ylabel("Frequency")
7 plt.tight_layout()
8 plt.show()
9
10 pop_mean = iris["sepal_length"].mean()
11 pop_std = iris["sepal_length"].std(ddof=0)
12
13 print(f"(a) Population mean (mu)      = {pop_mean:.4f}")
14 print(f"(a) Population std (sigma)    = {pop_std:.4f}")
15

```



```

(a) Population mean (mu)      = 5.8433
(a) Population std (sigma)    = 0.8253

```

(a)-(b) Simulation steps

1. Create an empty vector (list) to store sample means.
2. Loop 1,000 times.
3. In each iteration, draw a random sample of size $n = 30$ from `sepal_length`.
4. Compute the sample mean and store it.

```

1 import numpy as np
2

```

```

3 rng = np.random.default_rng(42)
4
5 n = 30
6 n_iterations = 1000
7
8 # 1. Empty vector to store sample means
9 sample_means = []
10
11 # Loop 1000 times, drawing a sample of size n=30 and storing its mean
12 for _ in range(n_iterations):
13     sample = rng.choice(iris["sepal_length"], size=n, replace=False)
14     sample_means.append(sample.mean())
15
16 sample_means = np.array(sample_means)
17 print(f"Collected {len(sample_means)} sample means.")

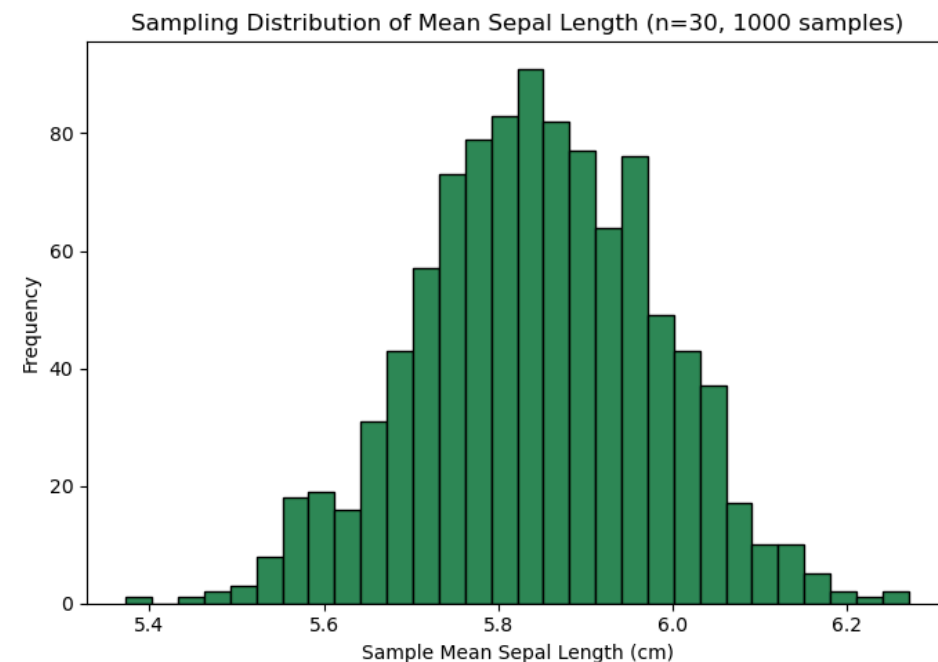
```

Collected 1000 sample means.

```

1 # (c) Histogram of the 1,000 sample means
2 plt.figure(figsize=(7, 5))
3 plt.hist(sample_means, bins=30, color="seagreen", edgecolor="black")
4 plt.title("Sampling Distribution of Mean Sepal Length (n=30, 1000 samples)")
5 plt.xlabel("Sample Mean Sepal Length (cm)")
6 plt.ylabel("Frequency")
7 plt.tight_layout()
8 plt.show()
9
10 # Empirical mean/std of the sampling distribution
11 mean_of_means = sample_means.mean() # type: ignore
12 std_of_means = sample_means.std(ddof=1) # type: ignore
13
14 # Theoretical values predicted by the CLT (using population mu/sigma from part a)
15 theoretical_se = pop_std / np.sqrt(n)
16
17 print(f"Empirical mean of sample means   = {mean_of_means:.4f}   (population mu = {pop_mean:.4f})")
18 print(f"Empirical std of sample means    = {std_of_means:.4f}   (theoretical sigma/sqrt(n) = {theoretical_se:.4f})")
19

```



```

Empirical mean of sample means   = 5.8444   (population mu = 5.8433)
Empirical std of sample means    = 0.1360   (theoretical sigma/sqrt(n) = 0.1507)

```

Discussion

The histogram of the 1,000 sample means is roughly bell-shaped and symmetric, even though individual `sepal_length` values are not perfectly Normal — exactly what the CLT predicts for $n = 30$. The empirical mean of the sample means is very close to the population mean μ , and the empirical standard deviation of the sample means closely matches the theoretical standard error $\sigma/\sqrt{n}$. This confirms the CLT: $\mu_{\bar{x}} \approx \mu$ and $\sigma_{\bar{x}} \approx \sigma/\sqrt{n}$, with the sampling distribution far more tightly clustered around the mean than the original population.

Question 6: Assessing a New Policy

Given *Normal distribution*:

$$X \sim N(\mu = 70,000, \sigma = 18,000)$$

z-score:

$$z = \frac{x - \mu}{\sigma}$$

a Percentage of households with income between 52,000 and 106,000

We want to find:

$$P(52,000 < X < 106,000)$$

Convert \$52,000 to a z-score

$$\begin{aligned} z_1 &= \frac{52,000 - 70,000}{18,000} \\ z_1 &= \frac{-18,000}{18,000} = -1 \end{aligned}$$

Convert \$106,000 to a z-score

$$\begin{aligned} z_2 &= \frac{106,000 - 70,000}{18,000} \\ z_2 &= 2 \end{aligned}$$

$$P(-1 < Z < 2)$$

From the standard normal table:

$$\begin{aligned} P(Z < 2) &= 0.9772 \\ P(Z < -1) &= 0.1587 \end{aligned}$$

Therefore:

$$\begin{aligned} P(-1 < Z < 2) &= 0.9772 - 0.1587 \\ &= 0.8185 \end{aligned}$$

Answer:

$$\boxed{81.85\%}$$

Approx.: **81.9% of households** have incomes between 52,000 and 106,000.

(b) Minimum income to be in the top 5%

"Top 5%" means **95% of households are below this income**.

then we need the **95th percentile**:

$$P(X < x) = 0.95$$

From the standard normal table:

$$z_{0.95} \approx 1.645$$

Then we can convert the z-score back to the original income:

$$x = \mu + z\sigma$$

$$x = 70,000 + (1.645)(18,000)$$

$$x = 70,000 + 29,610$$

$$x = 99,610$$

So a household needs to earn approximately **\$99,610 or more** to be in the top 5%.

(c) Maximum income to qualify for the bottom 20%

"Bottom 20%" means we need the **20th percentile**.

Thus we need to find the z-score for the 20th percentile:

$$P(X < x) = 0.20$$

Z score table:

$$z_{0.20} \approx -0.842$$

Convert this back to income:

$$x = \mu + z\sigma$$

$$x = 70,000 + (-0.842)(18,000)$$

$$x = 70,000 - 15,156$$

$$x = 54,844$$

So the maximum income is approximately \$54,844 to be in the bottom 20%.